THE NATIONAL UNIVERSITY OF LESOTHO

Department of Mathematics and Computer Science · Faculty of Science and Technology

M3503 — Ordinary Differential Equations

Mini-Project: One System, Many Methods

A semester-long modelling project

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1. The idea behind the project

Most students leave an ODE course able to execute the methods but unsure when or why to reach for each one. This project fixes that by giving every group one system of their choice and asking them to solve the same initial-value problem in three different ways: classical constant-coefficient methods, Laplace transforms, and a first-order system with eigenvalues. They then verify that all three agree.

The pedagogical pay-off is the outcome that is hardest to teach and assess in ordinary homework: **#H2 — compare different analytical methods**. When a group solves the same problem three separate ways and watches the answers coincide, the methods stop being isolated recipes and start being interchangeable tools, each with its own natural setting.

Because those methods are taught across the whole semester, the project is designed to **unfold unit by unit** rather than in a single burst, each part is released as the unit it depends on is covered (see Section 8), and it culminates in a compulsory group presentation.

2. Outcomes addressed

Course ELO	How the project covers it
#H1 solve IVPs / BVPs	Every part solves the same IVP under given initial conditions.
#H2 compare methods	Core of the project: three methods, one answer, explicit comparison.
#H3 systems of ODEs	Part D reformulates the model as a first-order linear system.
#H4 series methods	Part F (bonus) builds a power-series solution of a variable-coefficient variant.
#H5 real-life application	Every model is a real physical or biomedical system; Part E interprets its behaviour.
#S1–S2 problem-solving	Model derivation and regime classification from first principles.
#S3 teamwork	Carried out in groups of three to four, with a compulsory group presentation.

Course ELO	How the project covers it
#S4 time management	Staged across the semester, with a checkpoint as each relevant unit is completed.
#S5 technical report	A written report plus an oral presentation of the results.

3. Choose your system

Each group works with one system, chosen from the five below (or assigned, to spread groups across the range). The physics differs, but all five reduce to the same mathematical family — a damped second-order linear equation, or a 2×2 first-order linear system equivalent to one — so every model supports the full three-method comparison.

Model 1 — Mechanical: vehicle suspension

$$m\,x''(t) + c\,x'(t) + k\,x(t) = F_0\cos(\omega\,t)$$

A mass on a spring with a shock absorber, driven by a periodic road input. Given m, c, k, F_0, ω and the initial displacement and velocity. **Natural showcase:** forced oscillations and resonance.

Model 2 — Electrical: series RLC circuit

$$L\,q''(t) + R\,q'(t) + (1/C)\,q(t) = E(t), \quad E(t) = E_0\,u(t) \quad or \quad E_0\cos(\omega\,t)$$

The charge on the capacitor in a series RLC loop, driven by a source that is switched on at $t = 0$ (and, optionally, alternating). **Natural showcase:** Laplace transforms with the Heaviside step.

Model 3 — Environmental: two coupled mixing tanks

$$x_1' = -\alpha\,x_1 + \beta\,x_2 + f(t)$$

$$x_2' = \gamma\,x_1 - \delta\,x_2$$

Salt (or a pollutant) moving between two connected tanks under steady flow, with an inflow that switches on at $t = 0$. A first-order 2×2 system; reduce it to a single second-order ODE by elimination. **Natural showcase:** linear systems and the eigenvalue method.

Model 4 — Biomedical: two-compartment drug kinetics

$$c_b' = -(k_{10} + k_{12})\,c_b + k_{21}\,c_t + D(t)$$

$$c_t' = k_{12}\,c_b - k_{21}\,c_t$$

A drug exchanging between blood and tissue and clearing from the body, dosed as a single bolus (an initial condition) or a constant infusion (a step input). **Natural showcase:** systems and Laplace, with a clinically meaningful reading of the eigenvalues as clearance rates.

Model 5 — Electromechanical: DC motor speed response

$$L\,i' + R\,i + K_e\,\omega = V(t)$$

$$J\,\omega' + b\,\omega = K_t\,i$$

The current and angular speed of a DC motor after the supply voltage is switched on. Two coupled first-order equations that reduce to a single second-order ODE in the speed. **Natural showcase:** reducing a system to one ODE, and the step response.

Assign models — or, for the oscillator-type systems, different damping/eigenvalue regimes, so that groups' results complement one another and the class collectively sees the whole range.

4. The tasks

Part A — Build the model (#H5, #S1)

Starting from the governing physical law for your model (Newton's second law, Kirchhoff's voltage law, a mass balance, a rate law, or the motor equations), derive the equation(s). State the physical meaning of every term and parameter, and classify what you have (order, linearity, coefficients, homogeneity; for a system, its size and coupling).

Part B — Classical solution (#H1, #H2 · Unit 2)

Work with a single second-order ODE. If your model is a first-order system (Models 3–5), first **reduce it to one second-order equation by elimination** (Unit 4). Then:

1. Solve the homogeneous equation via the characteristic equation, and classify the roots (real distinct, repeated, or complex) — the over-/critically-/under-damped trichotomy or its analogue.
2. Find a particular solution to the input using undetermined coefficients.
3. Impose the initial conditions to obtain the full solution.

Part C — Laplace-transform solution (#H1, #H2 · Unit 3)

Solve the same IVP using Laplace transforms, with the input **switched on at t = 0** via a Heaviside step (so the second shift theorem earns its keep). Recover the solution by inverse transform and **confirm it matches Part B**. In one paragraph, explain why the transform handled the initial conditions (or the coupling, for a system) more gracefully.

Part D — Systems & eigenvalues (#H3 · Unit 4)

Write your model as a first-order linear system $\boldsymbol{x'} = A\,\boldsymbol{x} + \boldsymbol{f}(t)$ (Models 3–5 already are one; for Models 1–2 set $x_1 = x$, $x_2 = x'$). Find the eigenvalues of A and show they are **exactly the characteristic roots from Part B**. Solve the homogeneous system by the eigenvalue method and comment on what this correspondence reveals about the link between Units 2 and 4.

Part E — Qualitative & steady-state behaviour (#H2, #H5)

Study the long-run behaviour. For the driven oscillators (Models 1–2), treat the driving frequency ω as a dial: find the steady-state amplitude, locate the resonance peak, and describe how it flattens as damping grows. For the tank, drug, and motor models (3–5), find the equilibrium / steady state and show how the **slowest eigenvalue** governs how quickly it is reached (settling time, clearance, run-up).

Part F — Bonus: series solution (#H4 · Unit 5)

For extra credit, replace a constant coefficient with a time-varying one (e.g. a stiffness $k(t) = k_0(1 + \alpha t)$, a fouling tank rate, or a warming motor resistance), giving a variable-coefficient equation with no elementary closed form. Construct a **power-series solution about the ordinary point t = 0**, giving the recurrence relation and the first four non-zero terms.

Part G — Computational check (#H5)

Numerically integrate the original IVP (Python / SciPy, Octave, or even a spreadsheet or GeoGebra) and **overlay the numerical curve on your analytical solution**. Comment on the agreement and on any drift.

5. Deliverables

- **A technical report (#S5)** — the primary written deliverable, structured as in Section 6.
- **A group presentation (#S3 · compulsory)** — about 10 minutes plus questions, held before or during revision week. Each group presents its model and headline results (its damping or eigenvalue regime, its resonance or settling behaviour), so the class collectively sees all five model types and all three solution methods. It doubles as a class-wide revision session.

6. Report structure

1. Introduction — the physical system and the question being asked.
2. Model derivation (Part A).
3. Three solutions — Parts B, C and D, each in its own subsection.
4. Comparison — a single table showing the three solutions agree, plus a short discussion of when each method is the natural choice.
5. Qualitative / resonance analysis (Part E), with a plot.
6. Computational verification (Part G), with the overlay plot.
7. Bonus (Part F), if attempted.
8. Conclusion & reflection — one short paragraph per group member on what surprised them.
9. References — use the course textbooks (Nagle/Saff/Snider; Zill).

Suggested length: 8–12 pages. Submissions are run through Turnitin per course policy.

7. Assessment rubric

Criterion	ELOs	Weight
Correct model derivation & classification	H5, S1	10%
Classical solution (Part B)	H1	12%
Laplace solution + shift theorem (Part C)	H1	12%
Systems / eigenvalue solution & the connection (Part D)	H3	12%
Method comparison & discussion	**H2**	**15%**
Qualitative / resonance analysis (Part E)	H2, H5	8%
Computational verification (Part G)	H5	7%
Report quality: structure, clarity, referencing	S5	12%
Group presentation	**S3, S5**	**12%**
Bonus: series solution (Part F)	*H4*	*+10%*

The **comparison** thread (Parts B + C + D + the comparison ≈ 50%) remains the core of the marks; the compulsory presentation adds a further 12% for communicating it clearly.

8. How it runs across the semester

Rather than releasing the project in one burst, unlock each part as the unit it depends on is taught. This keeps the workload steady, lets each part reinforce material just after students meet it, and means nothing is assessed before it has been covered. The schedule below is anchored to units.

When	Release / milestone
After Unit 1	Choose a model; complete Part A (form and classify the equation).
After Unit 2	Part B for the second-order models (1–2); begin Part E (resonance) and Part G (computation).
After Unit 3	Part C — Laplace transforms.
After Unit 4	Part D — eigenvalues; for the system models (3–5), this is also where the Part B reduction and classical solve fall.
After Unit 5	Part F — the series bonus.
Final weeks	Assemble and submit the report; deliver the compulsory presentations before or during revision week.

- **Groups:** three to four students.
- **Checkpoints:** each unlocked part can be spot-checked as formative feedback at the end of its stage, so groups never fall far behind.
- **Weighting:** fold it into the existing 10% “assignments and quizzes” component, or run it as an enrichment activity worth a bonus toward that component.
- **Revision value:** landing the presentations at revision time turns them into a structured recap of every model type and all three methods.

9. Extensions & alternatives

Keen groups may propose a different linear system for approval, for instance a coupled two-mass, three-spring mechanical system (whose eigenvalues are the normal-mode frequencies), or a linearised SIR-type epidemic model about its equilibrium. Anything that stays within the course's linear framework and supports the three-method comparison is fair game.